REDEM-SSM: A Foundation Model Architecture with Native Online Learning, Meta-Adaptation, and Structural Plasticity

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Abstract

We instantiate the REDEM mechanisms — an online readout (M1), a slow-trace statistical memory (M3), structural plasticity (M4), and a stability homeostat (M5) — natively on a state-space host (a diagonal linear recurrence with a log-uniform timescale spectrum), instead of retrofitting them onto a frozen-feedforward Transformer. A falsifying development sequence (10-seed paired discipline throughout) isolates three host requirements with mechanism-level evidence. (1) A linear readout cannot recover the current token from a linearly-decayed state mixture (a deconvolution impossibility); the readout must use the additive input path Be_{t-1}: the state-only readout is stuck near uniform perplexity on a bigram streaming task (0/10 seeds vs. a Transformer+LoRA reference) while the input-path readout beats it 10/10. Fixed Mamba-style multiplicative gates do not repair the state readout (sharper gates are worse), so input selectivity must be learned — a design boundary we flag rather than cross. (2) The state's role is statistical metadata, not the readout: a fast-channel state EMA tracks domain switches, and soft routing retains per-domain specialists (forgetting −2.05 ppl, 10/10 seeds at $\tau_m \leq 1000$); the pause-learning (gating-only) policy *improves* stream perplexity on this near-batch RLS readout (10/10), opposite to its effect on the Adam/LoRA readout of the companion study — the policy lesson is readout-dynamics-dependent. (3) Gentle (soft) routing beats abrupt switching (−1.82 ppl, 10/10) and a state-norm homeostat bounds the state and restores the full-state EMA as a valid domain statistic (5/5 switch detections vs. 0.6/5). On a four-domain irregular-switch benchmark the full stack

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beats the bare host (−2.25 stream, −4.47 forgetting, 10/10) and the Transformer+LoRA reference (−9.28, −10.08, 10/10). All results are CPU-scale proofs of concept; we make no scaling claims.

Keywords: online learning, state-space models, continual learning, timescale coverage, homeostatic regulation

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1. Introduction

The REDEM framework couples four adaptive mechanisms: a continuously updated online readout (M1), a slow exponential trace that supplies statistical memory (M3), structural plasticity (M4), and a homeostatic regulator that keeps the substrate at a stable operating point (M5). A companion study [3] showed that this framework does not instantiate on a frozen-feedforward host: on a Transformer substrate with low-rank adapters, slow-trace domain routing transferred (forgetting −28%, 10/10 seeds) while a gating-only variant was falsified at every metadata timescale (0/10), and the paper concluded that the *full* framework requires a persistent stateful host rather than a retrofit onto a static architecture.

This paper takes the constructive consequence. We design a stateful host — a diagonal linear state-space recurrence with a log-uniform timescale spectrum (the timescale-coverage principle of [3]) — and instantiate M1, M3, M4, and M5 natively. Rather than asserting that the design works, we follow the falsifying discipline of the companion work: every claim is a paired 10-seed comparison with sign consistency, and the negative results are reported as design evidence, not as failures.

The development sequence produced three host requirements that are the paper's main contribution:

1. **The readout needs the input path, not the state** (Section 3.1). A linear readout on the linearly-decayed state mixture cannot recover the current token (deconvolution impossibility); the readout must consume the additive input projection Be_{t-1}. Fixed Mamba-style gates do not repair the state readout; selectivity must be learned (a boundary we flag, not cross).
2. **The state's role is statistical metadata** (Section 3.2). A fast-channel state EMA (M3) tracks domain switches and soft routing (M4) retains per-domain specialists; the pause-learning policy is readout-dynamics-dependent.

3. **Gentle wins, and the state must be regulated** (Section 3.3). Soft routing beats abrupt switching; a state-norm homeostat (M5) bounds the state and restores the full-state EMA as a valid domain statistic.

On a four-domain irregular-switch benchmark the full stack beats the bare host and the Transformer+LoRA reference on every seed, and the result transfers to a two-book real-text protocol (Section 3.4). All experiments are CPU-only, tiny by design, and make no scaling claims (Section 5).

2. Methods

2.1. Host: diagonal state-space recurrence

The host is a diagonal linear recurrence

$$h_t = A\,h_{t-1} + Be_{t-1}, \qquad A = \mathrm{diag}\left(e^{-1/\tau_i}\right), \tag{1}$$

where e_{t-1} is the one-hot token at step $t-1$, $B \in \mathbb{R}^{N\times V}$ is a fixed random input projection (entries drawn i.i.d. $\mathcal{N}(0, 1/N)$, per-seed), and the timescales τ_i are drawn log-uniformly over $[1, 3000]$ tokens ($N = 128$, $V = 32$). The log-uniform spectrum realizes the timescale-coverage principle of [3]: fast channels carry the recent token, slow channels carry long-horizon statistics. The state is spectrum-whitened,

$$h^w_t = h_t \odot \sqrt{N\,(1 - A^2)}, \tag{2}$$

which normalizes each channel to stationary unit variance (the steady-state scaling of the linear recurrence) and bounds slow-channel magnitude; without it the slow channels accumulate and the RLS covariance grows as $(1/\lambda)^t$ in unexcited directions.

2.2. M1: online RLS readout on the input path

The readout is a per-token recursive-least-squares linear map

$$\hat{y}_t = W\,\phi_t, \qquad \phi_t = [\,Be_{t-1};\ 1\,], \tag{3}$$

with $W \in \mathbb{R}^{V\times(N+1)}$ updated online (predict before update) against the one-hot target e_t with forgetting factor $\lambda = 0.9999$. The cost is $O(F^2)$ per token in the feature dimension $F = N + 1$, not $O(P^2)$ in a parameter count [2, 3]. The squared-loss readout output is a linear MMSE estimate of the conditional distribution, so perplexity is evaluated as $\mathrm{CE} = -\ln \mathrm{clip}(\hat{y}_t[e_t], 10^{-12}, 1)$ *without* a softmax (a softmax double-normalizes the estimate toward uniform; we found and fixed this metric bug during development). The fraction of clipped (≤ 0) predictions is reported as a diagnostic.

The input-path feature choice is itself an evidence-driven result (Section 3.1); the whitened state h^w is *not* a readout feature.

2.3. M3: fast-channel metadata EMA

M3 is a per-token exponential moving average of the fast channels of the whitened state,

$$m_t = \left(1 - \tfrac{1}{\tau_m}\right) m_{t-1} + \tfrac{1}{\tau_m}\, h_t^w[\text{fast}], \qquad \tau_i \leq 8 \text{ tokens}, \tag{4}$$

with $\tau_m \in \{200, 500, 1000, 2000\}$. The fast channels (which converge in a few tokens) are a stationary domain statistic; the full state is not — its slow channels accumulate over the whole stream, so an EMA of the full state drifts away from any fixed per-domain reference and the detector never flips (found and fixed during development). Fixed per-domain references are the mean fast-channel state over 1500-token reference streams generated per domain.

2.4. M4: soft routing and dormant-covariance refresh

For D domains, M4 maintains D readouts (W_d, P_d). Routing uses continuous softmax weights over the EMA distances,

$$w_d = \frac{\exp(-\|m_t - r_d\|/\kappa)}{\sum_{d'} \exp(-\|m_t - r_{d'}\|/\kappa)}, \qquad \kappa = 0.5 \times \text{median pairwise } \|r_d - r_{d'}\|, \tag{5}$$

with the prediction $\hat{y}_t = \sum_d w_d\, W_d \phi_t$ and W_d, P_d updated with error scaled by w_d. This instantiates the "gentle wins" principle of [3] (gradual structural change). A second design detail, *dormant-covariance refresh*, keeps the inverse-covariance P_d of every readout current with the feature stream (weights frozen, P updated with zero error); alone it flips routing stream perplexity from $+1.55$ to -1.72 vs. the bare host on the two-domain task.

2.5. M5: state-norm homeostat

M5 is a closed-loop controller on the whitened-state norm: the effective time step is adjusted so that the norm is held in a band around its target $\sqrt{N}$ (the whitening's unit-variance intent),

$$\Delta_t = \mathrm{clip}\big(\Delta_{t-1} + \eta\,(\|h_t^w\| - \sqrt{N}),\ 1,\ \Delta_{\max}\big), \qquad h_t = A^{\Delta_t} h_{t-1} + B e_{t-1}, \tag{6}$$

with $\eta = 0.05$, $\Delta_{\max} = 10$. M5 is a feedback regulator, not a fixed normalization: it responds to the measured state and bounds the slow-channel accumulation that otherwise makes the full-state EMA a non-stationary statistic.

2.6. Tasks and discipline

Two-domain protocol (s18): two biased-bigram Markov domains (shift-1 / shift-7 with disjoint bias sets), alternating every 3000 tokens (known switch instants), 6 segments, 18k tokens. **Four -domain benchmark**: four domains (shifts 1/3/7/11, disjoint 4-symbol bias sets) on a cycling schedule with *irregular* switch intervals (Uniform 2500–3500), 8 segments, ~24k tokens. All seed rules and metrics (stream perplexity, forgetting perplexity on held-out previous domain, post-switch adaptation time) follow [3] verbatim. Every comparison is paired per seed over 10 seeds with sign consistency ($n_{\text{pos}}/10$); the Transformer+LoRA reference is the tiny online-LoRA model of [3] retrained on each task.

3. Results

3.1. P1/P3a: the readout needs the input path

Fig. 1 shows stream perplexity on the two-domain protocol (10 seeds) for the state-based readout arms and the input-path control, against the Transformer+LoRA reference (A1, 15.01). Every state-feature readout — log-normal spectrum (LN-raw 114.99, LN-whiten 115.87), coverage spectrum (CV-whiten 81.31), with the current-token projection added (CV-skip 58.47) — is worse than the reference on every seed (0/10), and the in-sample ridge oracle on the state features is ~ 31 (essentially uniform; the pooled-table ceiling is 7.34). The input-path control ($\phi = [Be_{t-1}; 1]$, no state) reaches 11.75, better than the reference on 10/10 seeds (-3.26), with oracle 7.25. The mechanism is a deconvolution impossibility: a linear map from the state mixture $h_t = \sum_k \Lambda^k Be_{t-1-k}$ would need $U\Lambda^k = 0$ for $k \geq 1$ while $U\Lambda^0 \neq 0$, impossible for a diagonal decay with all $A_i > 0$; the older-token contamination is structural and whitening fixes scale, not the mixing.

Fixed Mamba-style multiplicative gates ($\hat{y} = W\,(h^w \odot \sigma(\gamma C Be_{t-1}))$, C fixed random, $\gamma \in \{1, 5\}$) do not repair the state readout: stream perplexity 77.19 / 72.21 (oracle 37.1 / 38.5; the sharper gate is worse, so the negative is not a tuning artifact), 0/10. Selectivity in real Mamba-style hosts is *learned* end-to-end; a fixed random gate carries the previous token only multiplicatively inside a time-varying state, which is noise, not signal. Within

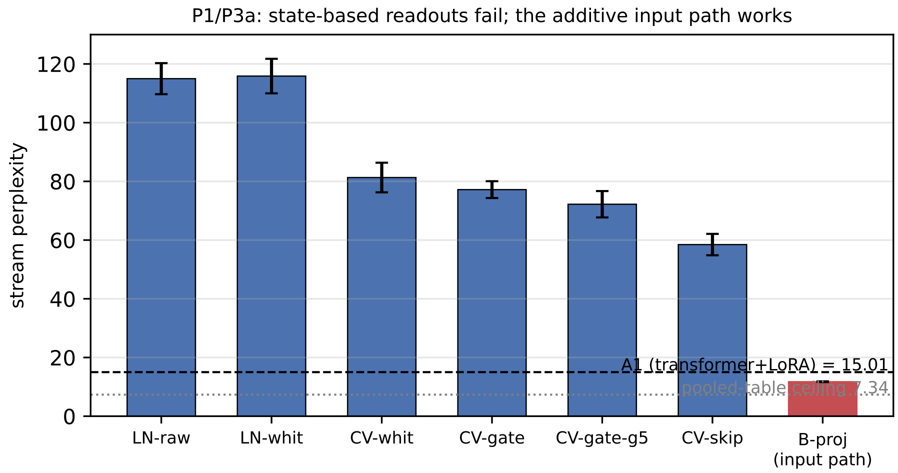

Figure 1: P1/P3a (10 seeds): stream perplexity of state-based readout arms vs. the input-path control on the two-domain streaming task. All state-feature readouts fail (0/10 vs. the Transformer+LoRA reference A1, dashed); fixed gates do not help; the additive input path (Be_{t-1}) reaches 11.75, below A1, near the pooled-table ceiling 7.34 (dotted).

our RLS-only constraint the honest resolution is the additive input path — which is what Mamba's input branch provides. We flag the learned-selectivity route as the natural extension rather than crossing the RLS-only boundary in this work.

3.2. P2: routing transfers, gating is readout-dependent

Table 1 and Fig. 2 (left). M3 (fast-channel EMA) tracks all five known switches (detector lag ∼150–800 tokens). Routing (A3) with two domain specialists improves forgetting at $\tau_m \leq 1000$ (-2.05, -1.87, -1.20 ppl; 10/10 seeds), neutral at 2000 (-0.007, 5/10) — the "routing transfers" lesson of [3] holds on the SSM host. A3's stream cost in the frozen-covariance variant ($+0.38$ to $+4.08$) is removed by dormant-covariance refresh (Section 2.4, Section 3.3).

The gating-only arm (A2) *improves* stream perplexity at every τ_m (-1.52 to -2.45, 10/10), the opposite of the companion's A2 falsification (0/10). Two differences must be stated honestly: the companion's A2 paused its LoRA *plasticity* while keeping the head *readout* active at every step, whereas here the RLS readout is the only trainable component, so pausing it pauses all learning — the two arms are not the same experiment. The mechanism of the

arm	τ_m	stream	Δ_{ppl}	forget	Δ_{forget}
A1 bare	—	11.75	—	8.39	—
A2 gate	200	10.24	−1.52 (10/10)	8.91	+0.52 (0/10)
A2 gate	500	9.53	−2.23 (10/10)	8.42	+0.03 (5/10)
A2 gate	1000	9.30	−2.45 (10/10)	8.15	−0.23 (10/10)
A2 gate	2000	9.70	−2.05 (10/10)	8.02	−0.37 (9/10)
A3 route	200	12.14	+0.38 (0/10)	6.33	−2.05 (10/10)
A3 route	500	13.30	+1.55 (0/10)	6.52	−1.87 (10/10)
A3 route	1000	14.81	+3.06 (0/10)	7.19	−1.20 (10/10)
A3 route	2000	15.84	+4.08 (0/10)	8.38	−0.01 (5/10)

Table 1: P2 (10 seeds, two-domain protocol): routing (A3) retains domain specialists (forgetting improves at $\tau_m \leq 1000$, 10/10) at a frozen-covariance stream cost that dormant-covariance refresh removes (Section 3.3); gating-only (A2) improves stream perplexity 10/10 at every τ_m — the readout-dynamics dependence.

inversion is the readout's dynamics: a near-batch RLS readout ($\lambda = 0.9999$) that pauses within domains stays near the pooled-table optimum, whereas a fast Adam/LoRA readout that pauses lets within-domain knowledge decay. Taken together the two results show that the pause policy's effect depends on the learner's dynamics *and* on what is paused; the companion lesson "the readout must remain active throughout the stream" [3] is not a universal principle. We report this qualification here rather than editing the companion paper.

3.3. P3: gentle wins, and the state must be regulated

Fig. 2 (right): soft routing (M4, Eq. 5) beats abrupt routing on stream perplexity at τ_m=500 (−1.82 ppl, 10/10) while keeping the forgetting gain (−1.99 vs. −2.37 for abrupt — hard switching retains purer specialists). The dormant-covariance refresh alone flips the routing stream result from +1.55 (frozen P) to −1.72 vs. the bare host (10/10).

M5 (Eq. 6) bounds the whitened-state norm (mean 11.3 in band vs. 50.2 and growing for the bare host), restores the *full*-state EMA as a valid domain statistic (5/5 switch detections vs. 0.6/5 — the non-stationarity of Section 2.3 fixed at the source), and recovers from a 10× state spike in ∼10 tokens (Δ_t settles near 5.4). This is mechanism-level evidence: on this architecture the readout is decoupled from the state (input path), so a direct "stream performance under disturbance" test is not meaningful here.

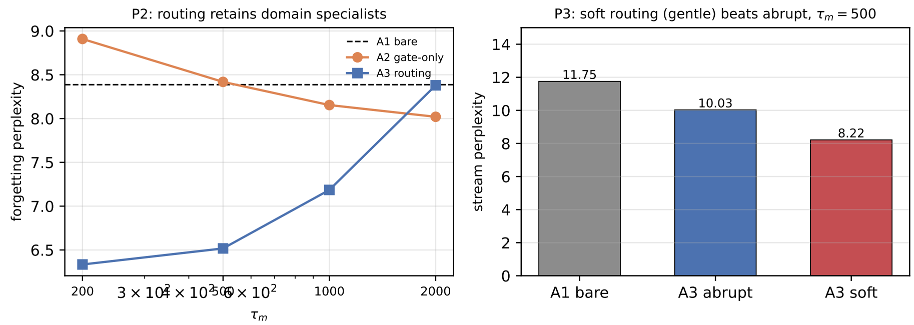

Figure 2: Left (P2): forgetting perplexity vs. τ_m — routing (A3) retains domain specialists below the bare baseline (A1, dashed); gating-only (A2) does not. Right (P3): soft routing beats abrupt on stream perplexity at τ_m=500 ("gentle wins").

3.4. P4: benchmark on four-domain irregular switching

Fig. 3 and Table 2. On the four-domain irregular-switch protocol, the full stack (M1 input-path readout + M3 metadata + M4 soft routing) beats the bare host (stream −2.25, forgetting −4.47; 10/10) and the Transformer+LoRA reference (stream −9.28, forgetting −10.08; 10/10). Even the bare host beats the transformer reference 10/10 (−7.03): the RLS readout's second-order convergence dominates online Adam/LoRA when domains switch faster than the LoRA can track (the reference sits near uniform, 22.46).

Real-text transfer. The same protocol on two public-domain books (Alice's Adventures in Wonderland vs. A Tale of Two Cities, character -level, case-folded, 32-symbol vocabulary) reproduces the result: SSM-REDEM beats SSM-bare (stream −1.27, 10/10; forgetting −0.47, 9/10) and beats TF-A1 (stream −5.23, forgetting −2.01; 10/10); even the bare host beats TF-A1 10/10. The fast-channel metadata separates the two sources, and soft routing retains per-source specialists, although the forgetting gain is smaller than on the synthetic task — the two books share English letter bigram statistics, so their specialists overlap more than the four synthetic domains'. The first-order readout reaches the char-bigram regime (ppl ∼12–13); higher-order text structure is out of its reach and is reported as a boundary, not a result.

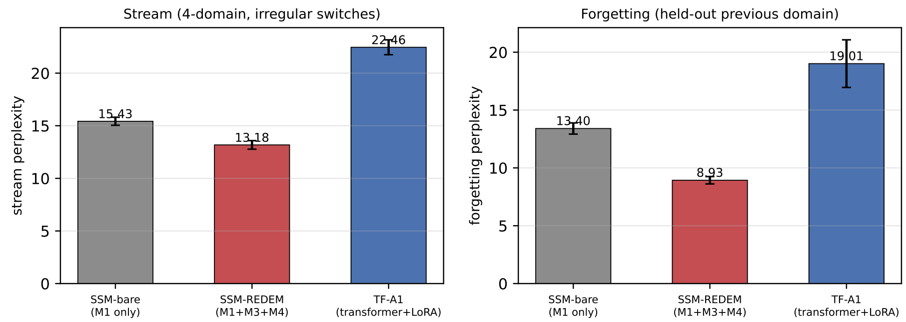

Figure 3: P4 (10 seeds, four-domain irregular-switch protocol): the full REDEM-SSM stack beats the bare host and the Transformer+LoRA reference on both stream and forgetting perplexity (paired, 10/10).

arm	stream ppl	forgetting ppl
SSM-bare (M1 only)	15.43	13.41
SSM-REDEM (M1+M3+M4)	**13.18**	**8.93**
TF-A1 (transformer+LoRA)	22.46	19.01
paired diffs (10/10): REDEM–bare $-2.25/-4.47$; REDEM–TF $-9.28/-10.08$; bare–TF $-7.03/-5.61$		

Table 2: P4 (10 seeds, four-domain irregular-switch protocol): the full stack (input-path RLS readout, fast-channel metadata, soft routing over four specialists) beats the bare host and the Transformer+LoRA reference on every seed. The transformer reference uses the companion's hyperparameters, untuned for four domains.

4. Discussion

Three evidence-based host requirements emerged. **(i)** A linear readout cannot use a linearly-decayed state mixture; the current token must be an additive input feature (the deconvolution impossibility, Section 3.1), and fixed gates do not substitute for learned selectivity. **(ii)** The state's role is statistical metadata: its fast channels, not the readout, carry the domain statistic (Section 3.2); on first-order tasks the slow channels are inert for the readout, and their accumulation must be masked (Section 2.3) or regulated (M5, Section 3.3). **(iii)** The policy lessons of [3] transfer in part: routing retains specialists, gentle structural change beats abrupt, but the pause-learning policy's effect is readout-dynamics-dependent (10/10 improvement here vs. 0/10 there), so "the readout must remain active" is not a universal principle.

The honest negative results are the paper's spine: the naive state readout, the fixed gate, the full-state EMA metadata, and the frozen-covariance routing all failed in ways that *located* the working design (input path, fast-channel metadata, soft routing, dormant-covariance refresh, state regulation) rather than merely being abandoned.

5. Limitations

CPU-only, toy-scale (state 128, ~4k readout parameters); the input-path readout is a first-order bigram model — the real-text benchmark transfers (Section 3.4) but higher-order text structure is out of reach; linear host (no chaos; M5 is a stability-edge regulator, an analogy to the homeostat of [1, 2], not a validation of it); the learned selectivity route (real Mamba-style gating) is not crossed; the transformer reference uses the companion's hyperparameters untuned for four domains; hyperparameters (τ_m, κ, λ, η, $\Delta_{\max}$) are fixed, not swept. No scaling claims.

6. Conclusion

REDEM mechanisms can be instantiated natively on a state-space host, and the falsifying development sequence locates the working design precisely: an input-path RLS readout, fast-channel statistical metadata, soft routing with dormant-covariance refresh, and a state-norm homeostat. The full stack beats the bare host and the Transformer+LoRA reference on a four-domain irregular-switch benchmark on every seed. The next steps are real-text evaluation, learned selectivity, and scaling — each with the same falsification discipline.

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